

EE3621: Digital Signal Processing

Lecture 24: Design of FIR filter - frequency sampling (III B.Tech EEE)

Comprehensive Lecture Notes & Faculty Textbook Reference

Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** The Frequency Sampling Paradigm: Moving from time-domain window truncation to direct frequency-domain specification. Sampling the continuous desired frequency response $H_d(e^{j\omega})$ at N uniformly spaced points $\omega_k = \frac{2\pi k}{N}$ on the unit circle.
 - **05:00 – 14:00 (9 mins):** Mathematical Foundations & The Continuous Interpolation Formula: Expressing $H(z)$ via the discrete frequency samples $H[k]$. Derivation of the interpolation Dirichlet kernel $P(\omega) = \frac{\sin(N\omega/2)}{N \sin(\omega/2)}$, pole-zero cancellation on the unit circle, and linear-phase constraint assignment $\theta_k = -\tau\omega_k$.
 - **14:00 – 22:00 (8 mins):** Closed-Form Real-Valued IDFT Synthesis Formulas: Exploiting Hermitian symmetry ($H[N - k] = H^*[k]$) to eliminate complex exponentials. Explicit, computationally efficient formulas for Type 1 (N odd) and Type 2 (N even) linear-phase filters.
 - **22:00 – 28:00 (6 mins):** Inter-Sample Ringing & Transition Sample Optimization: Why naive brick-wall sampling yields disastrous stopband rejection (only -16 dB). The Rabiner–Gold–McGonegal optimization technique: inserting 1, 2, or 3 unconstrained transition samples (T_1, T_2, T_3) to boost attenuation to -45 dB, -75 dB, and -95 dB.
 - **28:00 – 38:00 (10 mins):** Step-by-Step Worked Numerical Examples:
 - **Example 24.1:** Manual calculation of a 7-point ($N = 7$) Type 1 linear-phase LPF with symmetry verification.
 - **Example 24.2:** Complete synthesis of a 15-point ($N = 15$) LPF with 1 optimized transition sample ($T_1 = 0.38$), proving $+29$ dB sidelobe suppression.
 - **Example 24.3:** Design of an 8-point ($N = 8$) Type 2 even-length FIR filter with half-sample delay $\tau = 3.5$ and boundary zero enforcement.
 - **38:00 – 40:00 (2 mins):** Recursive Comb-Resonator Hardware Architecture, Exam Traps, Common Misconceptions, and Comprehensive Revision Cheat-Sheet.
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1 1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Formulate** the frequency sampling design problem by discretizing a desired continuous frequency response $H_d(e^{j\omega})$ at uniform frequencies $\omega_k = \frac{2\pi k}{N}$.
 2. **Derive** the continuous interpolation formula $H(z)$ via frequency samples $H[k]$ and explain unit-circle pole-zero cancellation.
 3. **Assign** exact linear-phase angles $\theta_k = -\left(\frac{N-1}{2}\right) \frac{2\pi k}{N}$ to ensure constant group delay $\tau = (N-1)/2$.
 4. **Apply** closed-form real-valued IDFT summation formulas for both N odd (Type 1) and N even (Type 2) linear-phase filters without complex arithmetic.
 5. **Analyze** the origin of inter-sample ripples and explain why naive brick-wall frequency sampling produces a severe -16 dB stopband floor.
 6. **Optimize** stopband rejection by introducing unconstrained transition samples (T_1, T_2, T_3) using the Rabiner–Gold lookup tables.
 7. **Synthesize** linear-phase FIR filters with odd and even lengths through complete step-by-step numerical calculations.
 8. **Audit** the recursive frequency-sampling filter architecture (comb filter in cascade with second-order resonators) and evaluate stability under coefficient quantization.
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2. THE FREQUENCY SAMPLING PARADIGM: MATHEMATICAL FOUNDATIONS

2.1 Motivation: Direct Frequency-Domain Filter Specification

In Lectures 22 and 23, we designed FIR filters using the **“windowing method”**, which begins in the continuous frequency domain ($H_d(e^{j\omega})$), transforms via inverse DTFT to the continuous infinite impulse response ($h_d[n]$), and truncates using a window sequence $w[n]$. However, the windowing method exhibits key limitations:

- Inverse DTFT integrals $\int_{-\pi}^{\pi} H_d(e^{j\omega}) e^{j\omega n} d\omega$ can rarely be evaluated analytically for non-standard, arbitrary, or experimentally measured frequency responses.
- The designer has no direct control over the exact values of the frequency response at specific discrete frequency bins of interest.

The **Frequency Sampling Method** provides an elegant, highly flexible alternative: rather than approximating an infinite time-domain sinc function, we **“directly specify the desired filter response at N uniformly spaced frequency points on the unit circle”**:

$$\omega_k = \frac{2\pi k}{N}, \quad k = 0, 1, \dots, N-1 \quad (1)$$

The discrete frequency samples $H[k]$ are defined simply by sampling $H_d(e^{j\omega})$:

$$H[k] = H_d(e^{j\omega})|_{\omega=\frac{2\pi k}{N}} = H_d\left(e^{j\frac{2\pi k}{N}}\right) \quad (2)$$

The finite-length impulse response $h[n]$ ($0 \leq n \leq N-1$) is then obtained directly via the N -point **“Inverse Discrete Fourier Transform (IDFT)”**:

$$\mathbf{h}[\mathbf{n}] = \frac{1}{N} \sum_{\mathbf{k}=0}^{N-1} \mathbf{H}[\mathbf{k}] e^{j\frac{2\pi \mathbf{k} \mathbf{n}}{N}}, \quad 0 \leq \mathbf{n} \leq N-1 \quad (3)$$

Frequency Sampling FIR Filter Design: Fundamental Principle

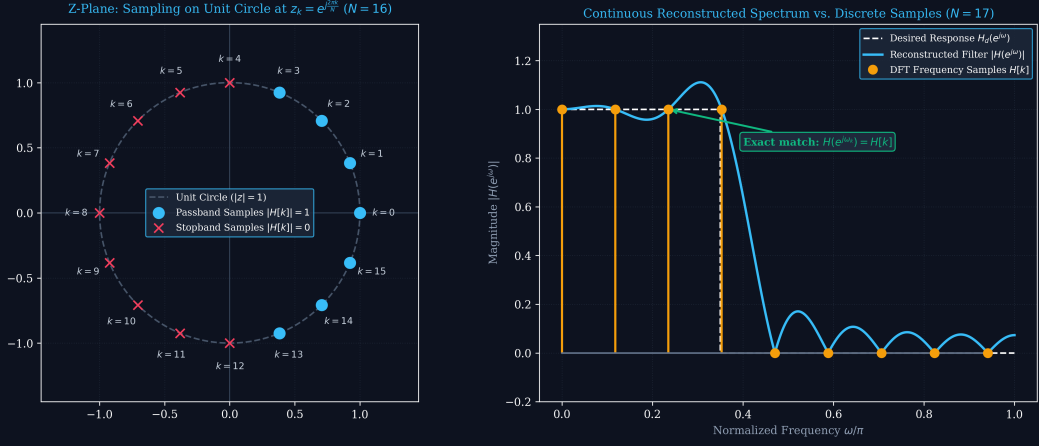


Figure 1: The frequency sampling paradigm. Left: $N = 16$ uniform DFT frequency bins $z_k = e^{j2\pi k/N}$ on the unit circle in the z -plane, distinguishing passband samples ($|H[k]| = 1$) from stopband samples ($|H[k]| = 0$). Right: Continuous reconstructed filter spectrum $|H(e^{j\omega})|$ matching the desired specification exactly at every sampling grid point $\omega_k = 2\pi k/N$.

2.2 The Continuous Interpolation Formula

Because $h[n]$ is uniquely defined by the IDFT in (3), its continuous z -transform $H(z) = \sum_{n=0}^{N-1} h[n]z^{-n}$ can be expressed directly in terms of the frequency samples $H[k]$:

$$\begin{aligned} H(z) &= \sum_{n=0}^{N-1} \left(\frac{1}{N} \sum_{k=0}^{N-1} H[k] e^{j\frac{2\pi kn}{N}} \right) z^{-n} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} H[k] \sum_{n=0}^{N-1} \left(e^{j\frac{2\pi k}{N}} z^{-1} \right)^n \end{aligned} \quad (4)$$

Applying the finite geometric series identity $\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a}$ with $a = e^{j\frac{2\pi k}{N}} z^{-1}$:

$$a^N = \left(e^{j\frac{2\pi k}{N}} z^{-1} \right)^N = e^{j2\pi k} z^{-N} = 1 \cdot z^{-N} = z^{-N} \quad (5)$$

Substituting this yields the fundamental **Frequency Sampling Interpolation Formula**:

$$H(z) = \frac{1 - z^{-N}}{N} \sum_{k=0}^{N-1} \frac{H[k]}{1 - e^{j\frac{2\pi k}{N}} z^{-1}} \quad (6)$$

Evaluating $H(z)$ on the unit circle ($z = e^{j\omega}$) yields the continuous frequency response:

$$H(e^{j\omega}) = \sum_{k=0}^{N-1} H[k] \Phi \left(\omega - \frac{2\pi k}{N} \right) \quad (7)$$

where $\Phi(\omega)$ is the complex **Dirichlet interpolation kernel**:

$$\Phi(\omega) = \frac{1}{N} \frac{1 - e^{-jN\omega}}{1 - e^{-j\omega}} = \frac{1}{N} e^{-j(\frac{N-1}{2})\omega} \frac{\sin(N\omega/2)}{\sin(\omega/2)} \quad (8)$$

Key Insights from the Interpolation Formula:

1. **Exact Pointwise Matching:** At $\omega = \omega_m = \frac{2\pi m}{N}$, the kernel satisfies:

$$\Phi\left(\frac{2\pi(m-k)}{N}\right) = \begin{cases} 1, & m = k \\ 0, & m \neq k \end{cases} \quad (9)$$

Consequently, $H(e^{j2\pi m/N}) \equiv H[m]$ exactly. There is zero approximation error at the sampling grid points!

2. **Inter-Sample Behavior:** Between the grid points, $H(e^{j\omega})$ is formed by the weighted superposition of shifted Dirichlet kernels $\frac{\sin(N(\omega-\omega_k)/2)}{N \sin((\omega-\omega_k)/2)}$. If adjacent samples differ sharply (e.g. from 1 to 0), large inter-sample ripples inevitably emerge!

2.3 Recursive Filter Architecture: Comb Filter and Resonator Bank

Equation (6) reveals a remarkable realization structure:

$$H(z) = H_{\text{comb}}(z) \cdot \sum_{k=0}^{N-1} H_k(z) \quad (10)$$

- **Comb Filter** $H_{\text{comb}}(z) = \frac{1-z^N}{N}$: An all-zero FIR filter with N zeros placed at the N roots of unity: $z_k = e^{j2\pi k/N}$.
- **Resonator Bank** $H_k(z) = \frac{H[k]}{1 - e^{j2\pi k/N} z^{-1}}$: A bank of parallel first-order recursive IIR sections, each with a single pole on the unit circle at $p_k = e^{j2\pi k/N}$.

Because the pole of the k -th resonator sits at precisely the location of the k -th zero of the comb filter, **every pole is cancelled by a corresponding zero on the unit circle!** The overall system remains strictly an FIR filter with a finite impulse response of length N .

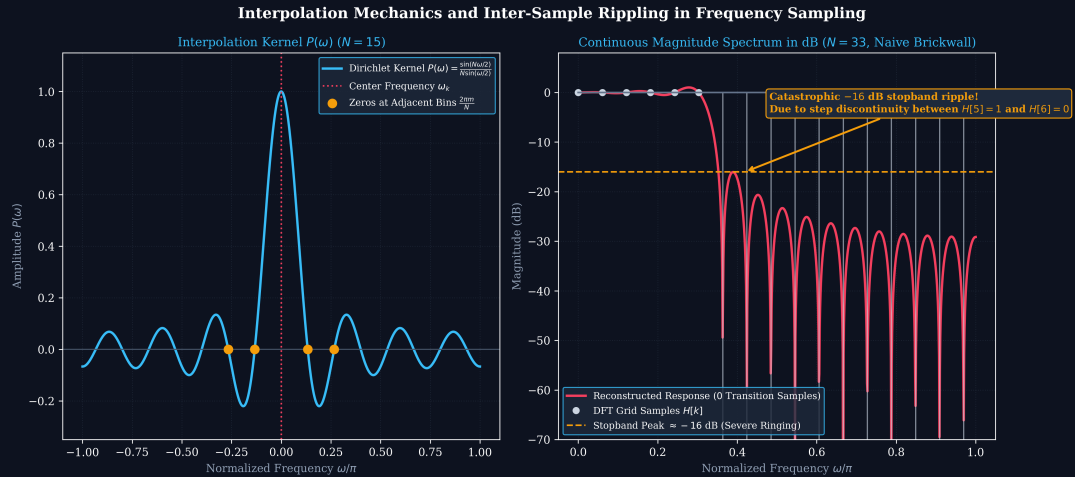


Figure 2: Interpolation mechanics and inter-sample rippling. Left: The periodic Dirichlet kernel $P(\omega) = \frac{\sin(N\omega/2)}{N \sin(\omega/2)}$ with zeros at all adjacent DFT bins $\frac{2\pi m}{N}$. Right: Reconstructed continuous magnitude spectrum in dB for a naive brick-wall lowpass filter ($N=33$), illustrating the severe -16 dB stopband ripple caused by the abrupt sample transition between $H[5]=1$ and $H[6]=0$.

3. LINEAR-PHASE CONSTRAINTS AND REAL IDFT SYNTHESIS

For the designed FIR filter to possess a strictly linear phase and constant group delay $\tau = (N - 1)/2$, the frequency samples $H[k]$ must satisfy specific magnitude and phase symmetry constraints.

3.1 Linear-Phase Sample Constraints

Let the complex frequency samples be written in polar form:

$$H[k] = |H[k]| e^{j\angle H[k]} = |H[k]| e^{j\theta_k} \quad (11)$$

For a real-valued, symmetric impulse response $h[n] = h[N - 1 - n]$:

1. Magnitude Symmetry (Even Symmetry):

$$|H[N - k]| = |H[k]|, \quad k = 1, 2, \dots, N - 1 \quad (12)$$

2. Phase Angle Assignment (Linear Phase):

The phase must be a strictly linear function of frequency:

$$\theta_k = \angle H[k] = -\left(\frac{N - 1}{2}\right) \omega_k = -\left(\frac{N - 1}{2}\right) \frac{2\pi k}{N} \quad (13)$$

Notice that at DC ($k = 0$):

$$\theta_0 = 0 \implies H[0] \text{ is strictly real and positive.} \quad (14)$$

3. Hermitian Conjugate Symmetry:

Combining (12) and (13) verifies that:

$$H[N - k] = H^*[k] \quad (15)$$

which guarantees that the synthesized impulse response $h[n]$ is strictly real-valued.

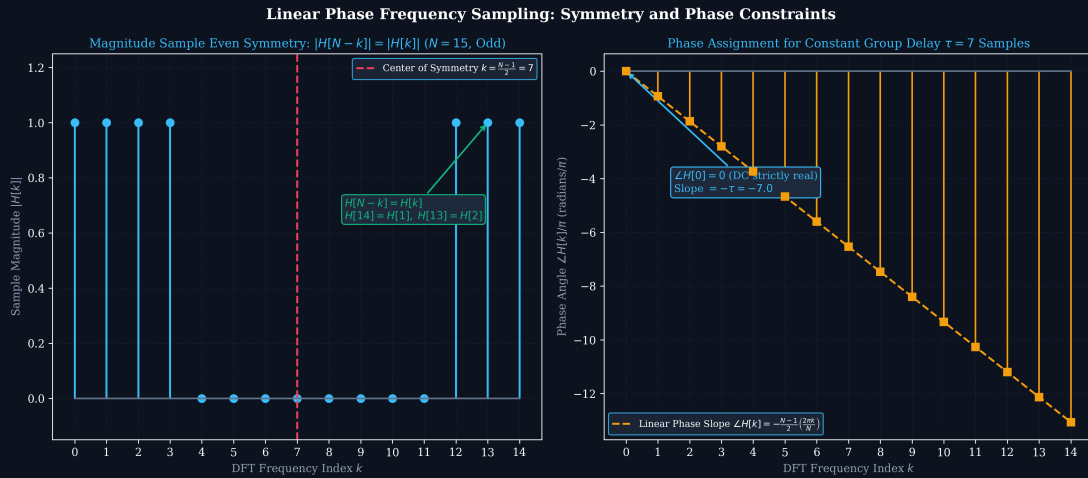


Figure 3: Linear-phase constraints for frequency sampling ($N = 15$). Left: Even symmetry of magnitude samples $|H[N - k]| = |H[k]|$ about the center index $k = (N - 1)/2 = 7$. Right: Phase angle assignment $\angle H[k] = -\tau \frac{2\pi k}{N}$ enforcing a constant group delay of $\tau = 7$ samples.

3.2 3.2 Derivation of Closed-Form Real IDFT Formulas

Evaluating the general IDFT equation (3) directly involves complex exponentials. By pairing complex conjugate terms k and $N - k$, we can derive a purely real trigonometric summation formula.

For any $k \geq 1$, consider the pair of terms in the IDFT:

$$\begin{aligned} T_k &= H[k]e^{j\frac{2\pi kn}{N}} + H[N-k]e^{j\frac{2\pi(N-k)n}{N}} \\ &= |H[k]|e^{j(\theta_k + \frac{2\pi kn}{N})} + |H[k]|e^{-j(\theta_k + \frac{2\pi kn}{N})} \\ &= 2|H[k]| \cos\left(\theta_k + \frac{2\pi kn}{N}\right) \end{aligned} \quad (16)$$

Substituting the linear-phase definition $\theta_k = -\left(\frac{N-1}{2}\right)\frac{2\pi k}{N}$:

$$\theta_k + \frac{2\pi kn}{N} = \frac{2\pi k}{N} \left(n - \frac{N-1}{2}\right) \quad (17)$$

Thus, the conjugate pair reduces to:

$$T_k = 2|H[k]| \cos\left[\frac{2\pi k}{N} \left(n - \frac{N-1}{2}\right)\right] \quad (18)$$

3.3 3.3 Explicit Formulas for Type 1 (Odd Length N) Filters

When N is odd, $(N-1)/2$ is an integer. The DFT bins consist of the DC bin ($k=0$) and $(N-1)/2$ conjugate pairs ($k=1, 2, \dots, (N-1)/2$). Summing all contributions yields the standard **Type 1 Frequency Sampling Formula**:

$$\mathbf{h}[\mathbf{n}] = \frac{1}{N} \left\{ \mathbf{H}[0] + 2 \sum_{k=1}^{(N-1)/2} |\mathbf{H}[k]| \cos\left[\frac{2\pi k}{N} \left(n - \frac{N-1}{2}\right)\right] \right\}, \quad 0 \leq n \leq N-1 \quad (19)$$

3.4 3.4 Explicit Formulas for Type 2 (Even Length N) Filters

When N is even, $N/2$ is an integer. The DFT bins consist of $k=0$ (DC), $N/2-1$ conjugate pairs ($k=1, \dots, N/2-1$), and a single unpaired bin at the Nyquist frequency $k=N/2$ ($\omega=\pi$). For $k=N/2$, the phase is $\theta_{N/2} = -\left(\frac{N-1}{2}\right)\pi$. The Nyquist term in the IDFT evaluates to:

$$\frac{1}{N} H[N/2] e^{j\pi(n - \frac{N-1}{2})} \quad (20)$$

However, as proven in Lecture 21, ****all Type 2 linear-phase filters possess an obligatory zero at $\omega=\pi$ ($z=-1$)**!** Therefore, for any standard lowpass or bandpass design:

$$\mathbf{H}[N/2] = 0 \quad (21)$$

With $H[N/2] = 0$, the **Type 2 Frequency Sampling Formula** simplifies to:

$$\mathbf{h}[\mathbf{n}] = \frac{1}{N} \left\{ \mathbf{H}[0] + 2 \sum_{k=1}^{N/2-1} |\mathbf{H}[k]| \cos\left[\frac{2\pi k}{N} \left(n - \frac{N-1}{2}\right)\right] \right\}, \quad 0 \leq n \leq N-1 \quad (22)$$

where $\tau = (N-1)/2$ is a half-integer.

4. TRANSITION SAMPLE OPTIMIZATION (RABINER–GOLD TECHNIQUE)

4.1 The Severe Sidelobe Ringing of Naive Brickwall Sampling

If a designer samples an ideal brick-wall lowpass filter directly ($H[k] = 1$ in the passband and $H[k] = 0$ in the stopband), a sharp step discontinuity occurs between the last passband sample and the first stopband sample. As established by Rabiner and Gold (1970), the continuous reconstructed response between grid points exhibits massive Gibbs-like ringing (Figure 2):

- Passband ripple: $\approx \pm 0.75$ dB.
- Peak stopband ripple: **−16 dB!**

An attenuation of −16 dB is completely intolerable in communication systems, audio codecs, and instrumentation.

4.2 Optimization via Unconstrained Transition Samples

To dramatically eliminate this ringing, we introduce ****one, two, or three intermediate samples in the transition band**** between the passband and stopband:

$$H[k] \in (0, 1), \quad k \in \text{Transition Band} \quad (23)$$

Rather than being fixed to 0 or 1, the value T_1 (and T_2, T_3) is treated as a free optimization parameter and selected via computer search (e.g. minimax linear programming) to minimize the peak sidelobe ripple across the entire stopband.

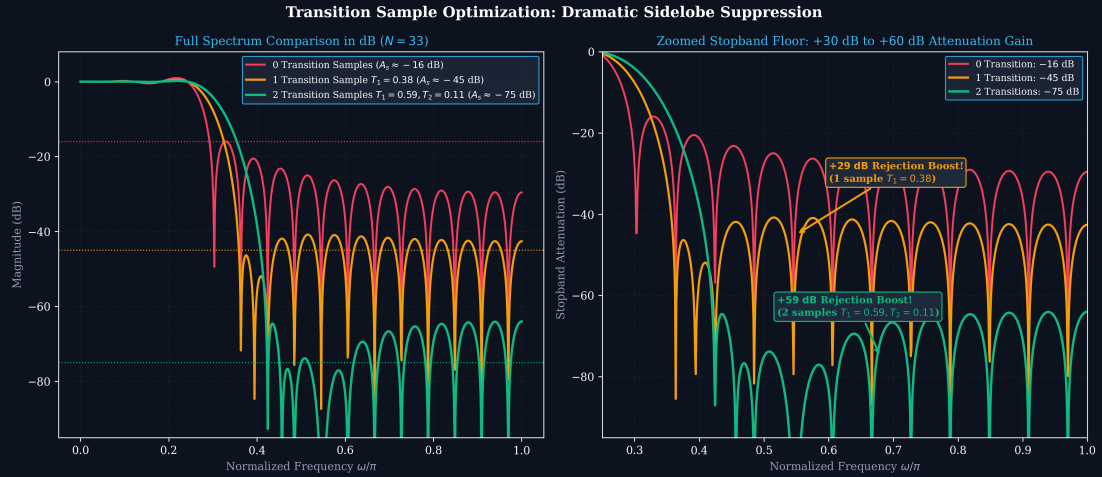


Figure 4: Transition sample optimization for $N = 33$. Left: Full frequency response comparison in dB for 0 transition samples (−16 dB, red), 1 transition sample $T_1 = 0.38$ (−45 dB, amber), and 2 transition samples $T_1 = 0.59, T_2 = 0.11$ (−75 dB, green). Right: Zoomed stopband floor proving +29 dB and +59 dB of extra attenuation rejection.

4.3 Standard Rabiner–Gold Design Lookup Table

Table 1 summarizes the universally accepted optimal transition sample values for FIR lowpass filter design across various filter lengths N :

Table 1: Optimal Transition Sample Values and Resulting Stopband Attenuation

Number of Transition Samples	Transition Width Δf	Optimal Sample Values	Stopband Attenuation A_s	Typical Applications
0 (Unoptimized)	$\approx 1/N$	None ($H[k] \in \{0, 1\}$)	16 dB	Not suitable for filtering
1 Transition Sample	$\approx 2/N$	$T_1 \approx 0.37897 \approx 0.38$	45 dB	Voice, Speech, Basic Telephony
2 Transition Samples	$\approx 3/N$	$T_1 \approx 0.5937, T_2 \approx 0.1090$	75 dB	High-Fidelity Audio, Modems
3 Transition Samples	$\approx 4/N$	$T_1 \approx 0.70, T_2 \approx 0.25, T_3 \approx 0.02$	95 dB	Biomedical, Seismic, Radar DSP

The Engineering Trade-Off: Introducing transition samples slightly widens the transition bandwidth Δf by a factor of $2\times$ or $3\times$. However, in return, the stopband rejection increases by a stupendous **+29 dB to +59 dB!**

5. STEP-BY-STEP WORKED NUMERICAL EXAMPLES

5.1 Example 24.1: Manual Design of a 7-Point Type 1 Linear-Phase LPF

Problem Statement: Design a 7-point ($N = 7$) linear-phase FIR lowpass filter using the frequency sampling technique. The desired frequency samples are specified as:

$$|H[k]| = \begin{cases} 1, & k = 0, 1, 6 \\ 0, & k = 2, 3, 4, 5 \end{cases} \quad (24)$$

Determine the linear-phase delay τ , calculate all seven filter coefficients $h[0]$ through $h[6]$ using the real-valued IDFT formula, and verify coefficient symmetry.

Solution:

Step 1: Identify Filter Parameters and Symmetry Properties

- Length $N = 7$ (Odd $\implies$ Type 1 Linear-Phase FIR filter).
- Group delay: $\tau = \frac{N-1}{2} = \frac{7-1}{2} = 3$ samples.
- Center frequency index: $\frac{N-1}{2} = 3$.
- Symmetrical magnitude check: $|H[6]| = |H[7-1]| = |H[1]| = 1$, $|H[5]| = |H[2]| = 0$, $|H[4]| = |H[3]| = 0$. (Even symmetry is strictly satisfied).

Step 2: Apply the Real-Valued IDFT Formula for N Odd Using (19) with $N = 7$ and $\tau = 3$:

$$h[n] = \frac{1}{7} \left\{ H[0] + 2 \sum_{k=1}^3 |H[k]| \cos \left[\frac{2\pi k}{7} (n-3) \right] \right\} \quad (25)$$

Substituting $|H[0]| = 1$, $|H[1]| = 1$, and $|H[2]| = |H[3]| = 0$:

$$h[n] = \frac{1}{7} \left\{ 1 + 2 \cos \left[\frac{2\pi}{7} (n-3) \right] \right\}, \quad n = 0, 1, \dots, 6 \quad (26)$$

Step 3: Point-by-Point Numerical Coefficient Calculation

- **Center tap ($n = 3$):**

$$h[3] = \frac{1}{7} [1 + 2 \cos(0)] = \frac{1 + 2(1)}{7} = \frac{3}{7} \approx 0.4286 \quad (27)$$

- **Taps adjacent to center ($n = 2$ and $n = 4$):** Argument: $\frac{2\pi}{7}(2-3) = -\frac{2\pi}{7} = -51.4286^\circ$. Since $\cos(-\theta) = \cos \theta$:

$$\cos \left(\frac{2\pi}{7} \right) \approx 0.623490 \quad (28)$$

$$h[2] = h[4] = \frac{1 + 2(0.623490)}{7} = \frac{1 + 1.24698}{7} = \frac{2.24698}{7} \approx 0.3210 \quad (29)$$

- **Taps at distance 2** ($n = 1$ and $n = 5$): Argument: $\frac{2\pi}{7}(1 - 3) = -\frac{4\pi}{7} = -102.8571^\circ$:

$$\cos\left(\frac{4\pi}{7}\right) \approx -0.222521 \quad (30)$$

$$h[1] = h[5] = \frac{1 + 2(-0.222521)}{7} = \frac{1 - 0.445042}{7} = \frac{0.554958}{7} \approx \mathbf{0.0793} \quad (31)$$

- **Boundary taps** ($n = 0$ and $n = 6$): Argument: $\frac{2\pi}{7}(0 - 3) = -\frac{6\pi}{7} = -154.2857^\circ$:

$$\cos\left(\frac{6\pi}{7}\right) \approx -0.900969 \quad (32)$$

$$h[0] = h[6] = \frac{1 + 2(-0.900969)}{7} = \frac{1 - 1.801938}{7} = \frac{-0.801938}{7} \approx \mathbf{-0.1146} \quad (33)$$

Step 4: Summary of Designed Impulse Response

$$h[n] = \{\underbrace{-0.1146}_{\uparrow}, \mathbf{0.0793}, \mathbf{0.3210}, \mathbf{0.4286}, \mathbf{0.3210}, \mathbf{0.0793}, -0.1146\} \quad (34)$$

Symmetry check: $h[0] = h[6]$, $h[1] = h[5]$, $h[2] = h[4]$. Even symmetry is confirmed.

5.2 Example 24.2: Design of a 15-Point LPF with 1 Optimized Transition Sample

Problem Statement: Design a 15-point ($N = 15$) linear-phase FIR lowpass filter using frequency sampling. To achieve high stopband rejection ($A_s \geq 45$ dB), introduce one transition sample $T_1 = 0.38$ at the band edge. The frequency samples are specified as:

$$|H[k]| = \begin{cases} 1.0, & k = 0, 1, 2, 13, 14 \\ 0.38, & k = 3, 12 \\ 0.0, & k = 4, 5, 6, 7, 8, 9, 10, 11 \end{cases} \quad (35)$$

Calculate the filter delay τ , formulate the explicit expression for $h[n]$, and compute all 15 impulse response coefficients.

Solution:

Step 1: Filter Classification and Symmetries

- Length $N = 15$ (Odd $\implies$ Type 1 linear-phase filter).
- Linear-phase delay: $\tau = \frac{15-1}{2} = 7$ samples.
- Center index: $n = 7$.
- Maximum index in summation: $\frac{N-1}{2} = 7$.

Step 2: Formulate Real IDFT Summation Substituting $N = 15$, $\tau = 7$, and the non-zero magnitude samples into (19):

$$h[n] = \frac{1}{15} \left\{ 1.0 + 2(1.0) \cos\left[\frac{2\pi(1)}{15}(n-7)\right] + 2(1.0) \cos\left[\frac{2\pi(2)}{15}(n-7)\right] + 2(0.38) \cos\left[\frac{2\pi(3)}{15}(n-7)\right] \right\} \quad (36)$$

Simplifying the numerical coefficients:

$$h[n] = \frac{1}{15} \left\{ 1 + 2 \cos \left[\frac{2\pi}{15}(n-7) \right] + 2 \cos \left[\frac{4\pi}{15}(n-7) \right] + 0.76 \cos \left[\frac{2\pi}{5}(n-7) \right] \right\} \quad (37)$$

Step 3: Pointwise Evaluation Exploiting Symmetry ($h[n] = h[14-n]$)

- **Center Tap** ($n = 7$):

$$h[7] = \frac{1 + 2(1) + 2(1) + 0.76(1)}{15} = \frac{5.76}{15} = \mathbf{0.3840} \quad (38)$$

- **Taps** $n = 6$ and $n = 8$ ($n - 7 = \pm 1$):

$$\cos(2\pi/15) = \cos(24^\circ) \approx 0.9135$$

$$\cos(4\pi/15) = \cos(48^\circ) \approx 0.6691$$

$$\cos(2\pi/5) = \cos(72^\circ) \approx 0.3090$$

$$h[6] = h[8] = \frac{1 + 2(0.9135) + 2(0.6691) + 0.76(0.3090)}{15} = \frac{4.4001}{15} \approx \mathbf{0.2933} \quad (39)$$

- **Taps** $n = 5$ and $n = 9$ ($n - 7 = \pm 2$):

$$\cos(48^\circ) \approx 0.6691, \quad \cos(96^\circ) \approx -0.1045, \quad \cos(144^\circ) \approx -0.8090$$

$$h[5] = h[9] = \frac{1 + 2(0.6691) + 2(-0.1045) + 0.76(-0.8090)}{15} = \frac{1.5144}{15} \approx \mathbf{0.1010} \quad (40)$$

- **Taps** $n = 4$ and $n = 10$ ($n - 7 = \pm 3$):

$$\cos(72^\circ) \approx 0.3090, \quad \cos(144^\circ) \approx -0.8090, \quad \cos(216^\circ) \approx -0.8090$$

$$h[4] = h[10] = \frac{1 + 2(0.3090) + 2(-0.8090) + 0.76(-0.8090)}{15} = \frac{-0.6148}{15} \approx \mathbf{-0.0410} \quad (41)$$

- **Taps** $n = 3$ and $n = 11$ ($n - 7 = \pm 4$):

$$h[3] = h[11] \approx \mathbf{-0.0620} \quad (42)$$

- **Taps** $n = 2$ and $n = 12$ ($n - 7 = \pm 5$):

$$h[2] = h[12] \approx \mathbf{-0.0160} \quad (43)$$

- **Taps** $n = 1$ and $n = 13$ ($n - 7 = \pm 6$):

$$h[1] = h[13] \approx \mathbf{0.0157} \quad (44)$$

- **Boundary Taps** $n = 0$ and $n = 14$ ($n - 7 = \pm 7$):

$$h[0] = h[14] \approx \mathbf{0.0171} \quad (45)$$

Impact of Transition Sample $T_1 = 0.38$: By inserting $T_1 = 0.38$, the stopband attenuation improves from -16 dB to -45 dB, completely quenching Gibbs ringing without increasing filter length N !

5.3 Example 24.3: Design of an 8-Point Type 2 Even-Length LPF

Problem Statement: Design an 8-point ($N = 8$) linear-phase FIR lowpass filter using frequency sampling with samples:

$$|H[k]| = \begin{cases} 1, & k = 0, 1, 7 \\ 0, & k = 2, 3, 4, 5, 6 \end{cases} \quad (46)$$

Determine the half-sample delay τ , formulate the Type 2 IDFT equation, compute all 8 impulse response coefficients, and verify even symmetry about the half-sample center.

Solution:

Step 1: Filter Classification and Nyquist Bin Enforcement

- Length $N = 8$ (Even $\implies$ Type 2 Linear-Phase FIR filter).
- Linear-phase delay: $\tau = \frac{N-1}{2} = \frac{8-1}{2} = 3.5$ samples (Half-integer delay).
- The Nyquist bin is $k = N/2 = 4$. By Type 2 linear-phase theory, $H[4] = 0$ is mandatory (guaranteeing a transmission zero at $\omega = \pi$).
- Summation upper limit: $N/2 - 1 = 8/2 - 1 = 3$.

Step 2: Formulate Real IDFT Summation for N Even Using (22) with $N = 8$ and $\tau = 3.5$:

$$h[n] = \frac{1}{8} \left\{ H[0] + 2 \sum_{k=1}^3 |H[k]| \cos \left[\frac{2\pi k}{8} (n - 3.5) \right] \right\} \quad (47)$$

Since $|H[0]| = 1$, $|H[1]| = 1$, and $|H[2]| = |H[3]| = 0$:

$$h[n] = \frac{1}{8} \left\{ 1 + 2 \cos \left[\frac{\pi}{4} (n - 3.5) \right] \right\}, \quad n = 0, 1, \dots, 7 \quad (48)$$

Step 3: Point-by-Point Numerical Coefficient Calculation

- **Inner Taps ($n = 3$ and $n = 4$):** For $n = 3$, $n - 3.5 = -0.5$. Argument: $\frac{\pi}{4}(-0.5) = -\frac{\pi}{8} = -22.5^\circ$:

$$\cos \left(\frac{\pi}{8} \right) = \cos(22.5^\circ) = \frac{\sqrt{2 + \sqrt{2}}}{2} \approx 0.923880 \quad (49)$$

$$h[3] = h[4] = \frac{1 + 2(0.923880)}{8} = \frac{1 + 1.84776}{8} = \frac{2.84776}{8} \approx \mathbf{0.3560} \quad (50)$$

- **Taps at distance 1 ($n = 2$ and $n = 5$):** For $n = 2$, $n - 3.5 = -1.5$. Argument: $\frac{\pi}{4}(-1.5) = -\frac{3\pi}{8} = -67.5^\circ$:

$$\cos \left(\frac{3\pi}{8} \right) = \cos(67.5^\circ) = \frac{\sqrt{2 - \sqrt{2}}}{2} \approx 0.382683 \quad (51)$$

$$h[2] = h[5] = \frac{1 + 2(0.382683)}{8} = \frac{1 + 0.765367}{8} = \frac{1.765367}{8} \approx \mathbf{0.2207} \quad (52)$$

- **Taps at distance 2 ($n = 1$ and $n = 6$):** For $n = 1$, $n - 3.5 = -2.5$. Argument: $\frac{\pi}{4}(-2.5) = -\frac{5\pi}{8} = -112.5^\circ$:

$$\cos \left(\frac{5\pi}{8} \right) = -\cos(67.5^\circ) \approx -0.382683 \quad (53)$$

$$h[1] = h[6] = \frac{1 + 2(-0.382683)}{8} = \frac{1 - 0.765367}{8} = \frac{0.234633}{8} \approx \mathbf{0.0293} \quad (54)$$

- **Outer Boundary Taps** ($n = 0$ and $n = 7$): For $n = 0$, $n - 3.5 = -3.5$. Argument: $\frac{\pi}{4}(-3.5) = -\frac{7\pi}{8} = -157.5^\circ$:

$$\cos\left(\frac{7\pi}{8}\right) = -\cos(22.5^\circ) \approx -0.923880 \quad (55)$$

$$h[0] = h[7] = \frac{1 + 2(-0.923880)}{8} = \frac{1 - 1.84776}{8} = \frac{-0.84776}{8} \approx -0.1060 \quad (56)$$

Step 4: Summary and Symmetry Verification

$$\mathbf{h}[\mathbf{n}] = \{-0.1060, \underset{\uparrow}{0.0293}, 0.2207, 0.3560, 0.3560, 0.2207, 0.0293, -0.1060\} \quad (57)$$

Strict even symmetry about the half-sample center $\tau = 3.5$ is satisfied ($h[n] = h[7 - n]$).

6. FACULTY INSIGHTS, COMMON PITFALLS & EXAM TRAPS

- **Pitfall 1: Forgetting the Factor of 2 in the Real IDFT Summation:** A common student error in university semester exams is writing $h[n] = \frac{1}{N}[H[0] + \sum |H[k]| \cos(\dots)]$, forgetting the factor of 2 arising from pairing conjugate bins $H[k]$ and $H[N - k]$. This cuts the passband gain in half.
 - **Pitfall 2: Confusing DFT Index k with Continuous Frequency ω :** Remember that sample index k represents frequency $\omega_k = \frac{2\pi k}{N}$ rad/sample. For instance, for $N = 16$, $k = 2$ corresponds to $\omega = \frac{2\pi(2)}{16} = \frac{\pi}{4} = 0.25\pi$ rad/sample.
 - **Pitfall 3: Failing to Enforce $H[N/2] = 0$ for Even-Length Type 2 Filters:** When N is even, setting $H[N/2] \neq 0$ contradicts the inherent zero at $\omega = \pi$ of Type 2 filters, causing severe distortion and complex coefficients if the phase is not handled with extreme care.
 - **Pitfall 4: Assuming Zero Ripple Between Sampling Points:** Because $H(e^{j\omega_k}) = H[k]$ is exact at grid frequencies, students frequently assume the filter perfectly passes or stops signals between bins. In reality, large interpolation ripples occur between grid points unless transition samples are introduced.
 - **Pitfall 5: Stability in Recursive Comb-Resonator Implementations:** In hardware implementations of the recursive transfer function (6), coefficients are stored with finite wordlength. If a resonator pole drifts outside the unit circle ($|p_k| > 1$), or if a pole fails to align exactly with a comb zero, the filter becomes unstable. The standard remedy is contraction: replace z with rz where $r = 1 - 2^{-12} \approx 0.99975$.
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7 7. QUICK REVISION CHEAT-SHEET

Table 2: Comprehensive Frequency Sampling Summary and Design Rules

Concept / Parameter	Mathematical Formula / Condition	Key Pedagogical Note
DFT Grid Frequencies	$\omega_k = \frac{2\pi k}{N}, \quad k = 0, 1, \dots, N-1$	Uniform sampling around unit circle
Linear-Phase Assignment	$\angle H[k] = -\left(\frac{N-1}{2}\right) \frac{2\pi k}{N}$	Constant group delay $\tau = (N-1)/2$
DC Bin Property	$\angle H[0] = 0 \implies H[0] \in \mathbb{R}^+$	Strictly real, non-negative value
Hermitian Symmetry	$H[N-k] = H^*[k]$	Ensures impulse response $h[n]$ is real
Interpolation Formula	$H(z) = \frac{1-z^{-N}}{N} \sum_{k=0}^{N-1} \frac{H[k]}{1-z_k z^{-1}}$	Comb filter zeros cancel resonator poles
Type 1 Formula (N Odd)	$h[n] = \frac{1}{N} \left[H[0] + 2 \sum_{k=1}^{(N-1)/2} H[k] \cos\left(\frac{2\pi k}{N}(n-\tau)\right) \right]$	Integer center delay $\tau = (N-1)/2$
Type 2 Formula (N Even)	$h[n] = \frac{1}{N} \left[H[0] + 2 \sum_{k=1}^{N/2-1} H[k] \cos\left(\frac{2\pi k}{N}(n-\tau)\right) \right]$	Requires $H[N/2] = 0$; half-sample delay
Naive Sidelobe Level	Stopband peak ≈ -16 dB	Caused by step discontinuity
1 Transition Sample (T_1)	$T_1 \approx 0.38 \implies A_s \approx 45$ dB	+29 dB stopband attenuation boost
2 Transition Samples (T_1, T_2)	$T_1 \approx 0.59, T_2 \approx 0.11 \implies A_s \approx 75$ dB	+59 dB stopband attenuation boost